* TIME SERIES FORECASTING *

Agenda →

Adv. Supervised Algos

Time series Analytics & forecasting

Recommender Systems

Netflix
Hotstar
Prime

Business → Product

Shipping → (use case) Service

Sensors → aggregate

inventory
Sales

Amazon
iphone
Diwali

Shipping
Airlines → TURN

kms

saven predictor

Agenda
- Prob. statement
- Missing values
- Anomalies
- Breaking down time series → Decomposition

Problem statement → DS

motiplus

forecast future sales for better planning & Revenue

- **Agenda 1:** We want to understand the patterns in demand to be able to be better plan for factory maintenance / staffing requirements.
- **Agenda 2:** We need a certain level of accuracy. The management requires that the Mean Absolute Percentage Error (MAPE) is not more than 5%.
- **Agenda 3:** Need a range forecast to supplement the point forecast to make educated trade-off wherever needed.

Sep 2026

600

MAPE → Next Class

MSE

R^2

Regression metrics

400 to 600

550

Agenda 1

Sales data

Why forecast ?

→ Planning

→ inventory

MoBiPlus → future sales →

* Sep → Dec

Diwali

(1) Resource management

* June July : sales ↓

Machines → maintenance

(2) Better planning

How much inventory do we want to keep

500 →

Requirement pred = 600 ↑ sep

sep → 480 ✓

overpredicted

350 →

overforecasted

(K) Underforecasting

(K) overforecast →

- dead inventory

- invested amount

underforecast

→ unmet demand

→ Revenue loss

domain

Pharma → cold → expiry

Dairy products → shelf life

→

Swiggy → morning → milk ✓

→ exhausted

evening ✗

Time SERIES DATA

t — day, date, hour, month, ... data (info)

	DATE	Sales
0	2001-01-01	6519.0
1	2001-02-01	6654.0
2	2001-03-01	7332.0
3	2001-04-01	7332.0
4	2001-05-01	8240.0

any signal indexed by ordered Timestamp

→ year, monthly, daily, hourly

t — sales

y_t = $f(t)$

y_{t+1}, y_{t+2}, y_{t+3}

June, July, Aug

time

Sales

Regression

Classification

Clustering

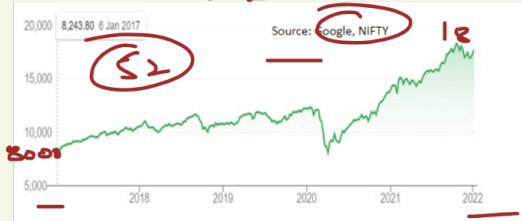
time

Advanced Regression

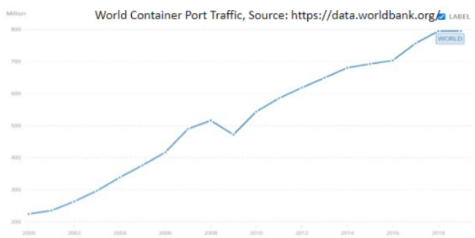
Sales

No i/p features

Time series









6-7 %.

4% → 4.5% +ve growth

→ 3.8% −ve

understand patterns

Forecasting →

inputs: y_1 y_2 y_3 ---- y_t

Aug 2026

output → $\hat{y}_{t+1}$ $\hat{y}_{t+2}$

Sep 26 Oct 26

Jan 2001 → Jan 2019 t+ t+2

ML → Supervised → Train test split

Shuffles

80%

⇒ 20%

Time series

Random splitting

X y

X_train y_train

X_test y_test

Aug Nov Sep

order ✓

y (sales)

DATE

80%

12 months → 20%

Testing

time → ✓

ordering →

CARS 24

year	price
2014	
2017	
2018	

Challenges

1. Missing sales in B/W
2. outliers

Quiz time!

Time Left: 55s

If the data contains only 3% of the missing values, dropping them is a good way of handling the missing data in a time series model.
Is the statement True or False?

1 user has participated

A True 100%

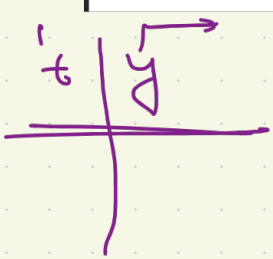
B False 0%

End Quiz Now

9:51

16:02

Time series



B PATEL → Registration

order

Supervised

imputation

KNN

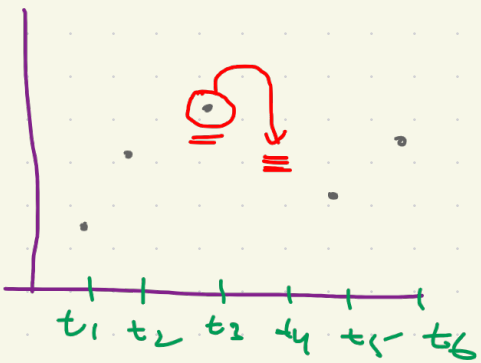
9

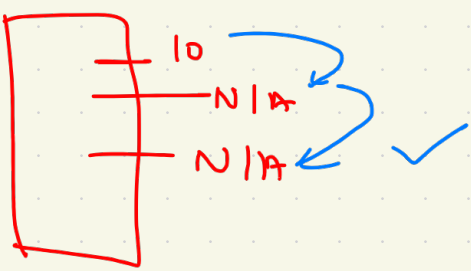
① → zero

② → mean / median

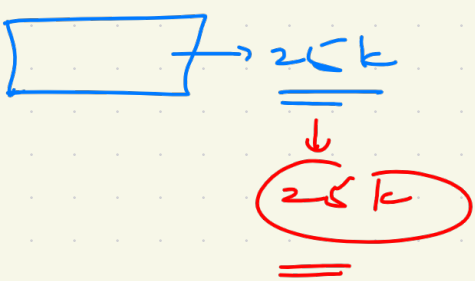
↳ ignores temporal locality

③ ffill



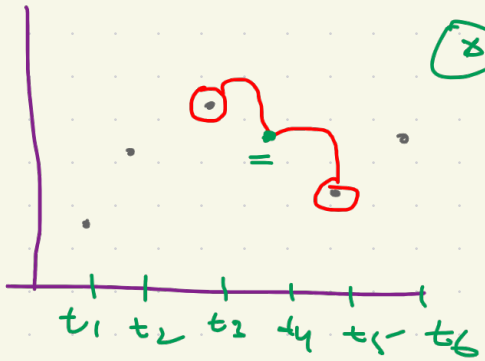


kirana



ffill

Interpolation



linear interpolation

missing

KNN

KNN

Similarity

ffill

Sales

find anomalies

Quiz time!

Time Left: 37s

Why is using mean values of the data not optimal for filling the missing values?

22 users have participated

A	The filled value may be different from local neighbours.	86%
B	As the time series has dates, it's not possible to find mean.	5%
C	Mean values will work fine even incase of missing data in time series.	9%

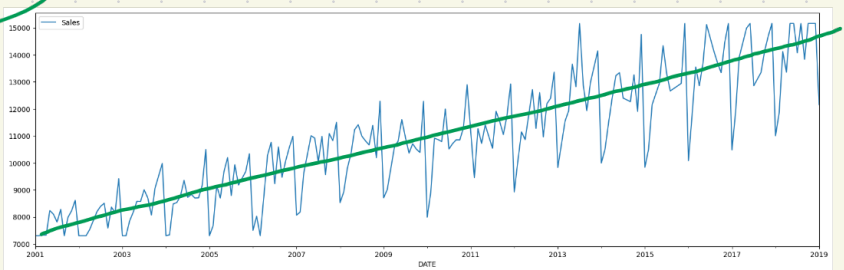
End Quiz Now

Components of Time Series

Trend

Seasonality

Business



+ve

May June July →

Sales ↑ → Diwali

↓

Business → Pattern

Pattern → monthly, yearly, quarterly

Repeating pattern = Seasonality



Observation :

Monday sales go down

Sales ↑ weekends

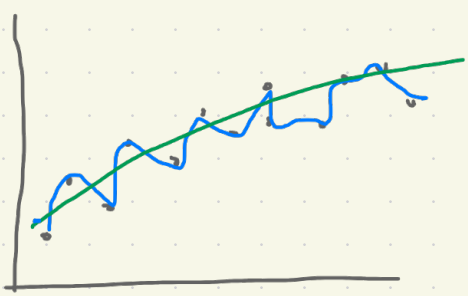
Diwali → ↑↑ → Next

→ previous

(1) Moving averages (MA)



→ identify trend



+ve growth

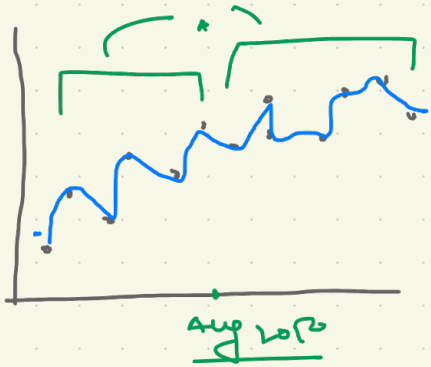
• rolling(3).mean()

```
mobile_sales['Sales'].rolling(3).mean()
```

DATE	Sales
2001-01-01	NaN
2001-02-01	NaN
2001-03-01	7315.866667
2001-04-01	7323.933333
2001-05-01	7634.666667
2001-06-01	7892.000000
2001-07-01	8052.333333
2001-08-01	8065.333333

x_1 x_2 x_3

Centred moving avg (CMA)



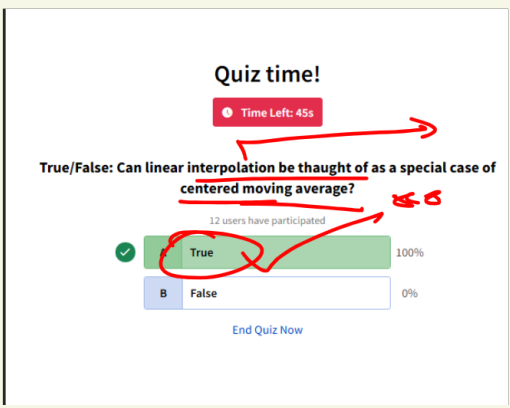
smoother the signal

Jan Feb Mar

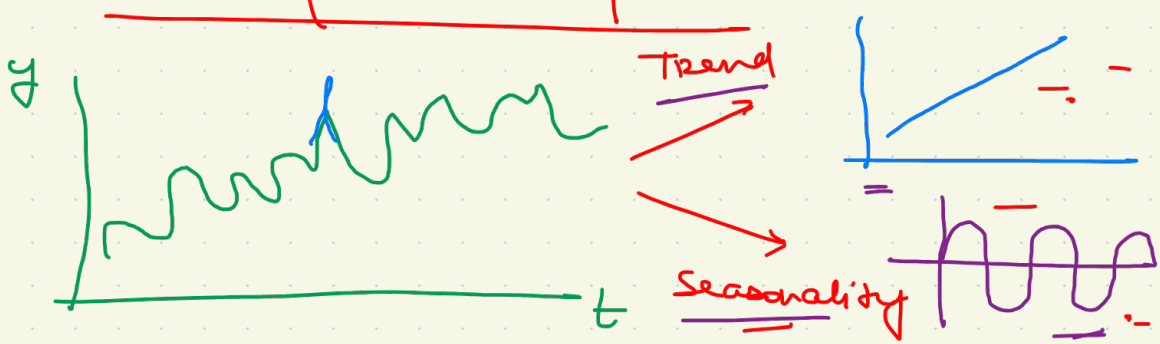
~~X~~ Feb Mar

mean

MA

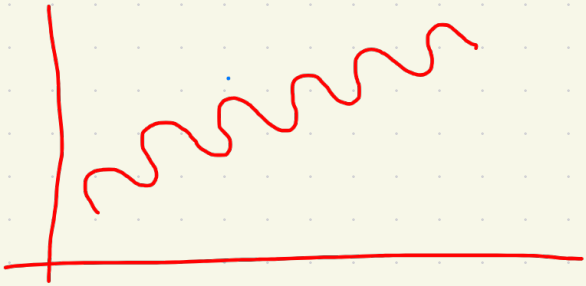


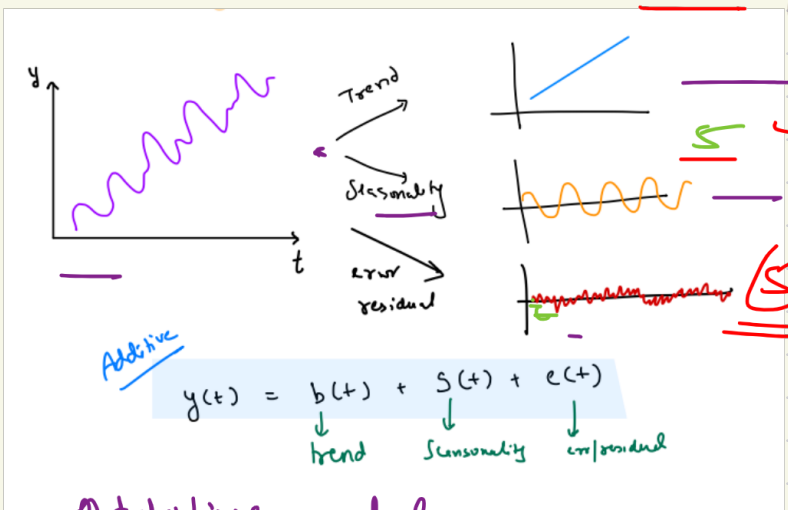
Decomposition of Time SERIES



$$y(t) = b(t) + s(t) + e(t)$$

error





10

20

S

S → residual

Additive model

$$\hat{y}(t)$$

$$e(t) = y(t) - [b(t) + s(t)]$$

actual sales

pred sales

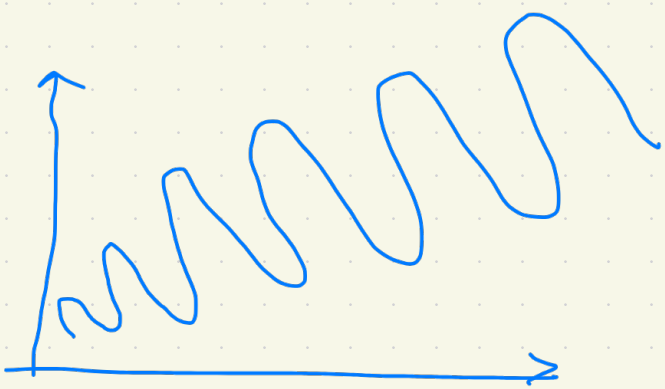
July → 15,400

t → T(t) = 14000
Trend

S(t) = 1000

14000 + 1000 = 15000

e(t) = 15400 − 15000
= 400 →



→ multiplicative model

Sales ↑

Variance ↑

$$y(t) = b(t) \cdot s(t) \cdot e(t)$$

$$e(t) = \frac{y(t)}{b(t) \cdot s(t)}$$

Done





